

EDUCATION

University of British Columbia

Masters of Engineering – Electrical and Computer Engineering Department

- Focused on Machine Learning and Software Engineering

University of British Columbia

Bachelors of Applied Science – Electrical and Computer Engineering Department

Completed: April, 2025

Grade: 4.0 GPA

Completed: April, 2024

Grade: 3.7 GPA

WORK EXPERIENCE

Quadrant Electronics Inc, North Vancouver, BC

Software, Firmware and ML Engineer

- Made UI and SQL Azure database improvements to custom inventory control software.
- Developed firmware in C++ and MicroPython in a Linux Docker environment for an ESP32 microcontroller that included functionality for I2C, SPI, Wi-Fi communications, and multithreading; conducted benchmark testing to validate and improve high-speed network communication.
- Modeled RF power amplifier impairments using various techniques including the development of a non-linear retentive neural network and a memoryless digital signal processing approach.

Apr, 2025 – Present

NeuroPrior AI, North Vancouver, BC

Software and Firmware Engineer

- Collaborated in a 5-person team to design a medical EEG/ECG wearable device.
- Developed firmware in C for BLE communication (advertising, connecting, ADC data transmission).
- Applied DSP to transform sensor input into real-time audio playback and FFT visualizations in both Python and MATLAB.
- Built a Python desktop application with a GUI to automate device connectivity, stream signals, and visualize data.

Apr, 2024 – Sep, 2024

Quadrant Electronics Inc, North Vancouver, BC

Junior Software Engineer

- Designed and prototyped a VR pilot training program in Unity, using C# scripts, with interactive cockpit controls and instrument simulations for fault-scenario training.
- Created an AR mobile application to extend training accessibility.
- Gained experience in 3D data interaction, rendering, and visualization.
- Engineered workflows independently from concept to working prototype, showcasing initiative and problem solving skills.

Apr, 2022 – Aug, 2022

Celly Technologies, North Vancouver, BC

Junior Software and Firmware Developer

- Worked in a 4-person team to design a mobile application that monitored sump pump status and provided real-time user alerts.
- Wrote an Android app using OpenCV for image/video frame processing, automating LED detection and event classification.
- Integrated AWS backend services to manage user data and make it accessible online, ensuring reliable and scalable data delivery.

Apr, 2020 – Aug, 2020

RESEARCH

Investigating the Role of Test Cases in LLM-Based Python and JavaScript Code Translation

- Investigated different prompt engineering and feature engineering approaches and iterative post processing techniques for two-way Python and JavaScript code translation. Our improvements increased the LLM translation success rate from 44% to 91%.
- Tests included generating synthetic test cases to translate code with different LLM models including Gemini-2.0 and GPT-3.5.

Vision Transformer Based Chessboard Image Analysis

- Explored the use of Vision Transformers for classifying digital chess board pieces for game state extraction.
- Involved extensive data pre-processing and was able to significantly reduce training complexity and time while maintaining a model generalizable to many board styles and colors. For 100% classification accuracy the training data required was reduced to 500 from 80,000.

Transformer Based Actor Critic Agent

- Investigated use cases of transformers in a reinforcement learning setting. Tests were done using an actor critic model.
- Utilized OpenAI Gym to test the model’s performance in different environments including Cart Pole and lunar Lander.
- Transformer model took longer to converge but was more stable. Reduced agent failure rate from 0.02% to 0.0%.

Languages		Tools			Software	
• Python	• C#	• Azure/AWS	• MSSQL/PostgreSQL	• OpenCV	• Unity	• Android Studio
• C and C++	• JavaScript	• Git	• Linux Docker	• PyTorch	• Matlab	• VSCode